

Solutions: Additional Exercises 5

Improper Integrals & Topology of $\mathbb{R}^n$

Exercise 1: Convergence or divergence

Reference: $\int_1^\infty x^{-p} dx$ converges iff $p > 1$; $\int_0^1 x^{-p} dx$ converges iff $p < 1$.

- (a) $\int_1^\infty \frac{dx}{x^3+1}$. For large x : $\frac{1}{x^3+1} \sim \frac{1}{x^3}$. Since $p = 3 > 1$, **converges** (limit comparison with $\int_1^\infty x^{-3} dx$).
- (b) $\int_0^1 x^{-1/3} dx$. Singular at 0, $p = 1/3 < 1$. **Converges**. Value: $\left[\frac{x^{2/3}}{2/3}\right]_0^1 = 3/2$.
- (c) $\int_1^\infty \frac{\ln x}{x^2} dx$. Integrate by parts: $u = \ln x$, $dv = x^{-2} dx$. $= -\frac{\ln x}{x} \Big|_1^\infty + \int_1^\infty \frac{1}{x^2} dx = 0 + \left[-\frac{1}{x}\right]_1^\infty = 1$. **Converges; value = 1**.
- (d) $\int_0^1 \frac{\sin x}{\sqrt{x}} dx$. Near 0: $\sin x \sim x$, so $\frac{\sin x}{\sqrt{x}} \sim x^{1/2}$. Since $p = -1/2 < 1$ (equivalently the function is bounded and integrable), **converges**.
- (e) $\int_0^\infty e^{-x} dx = [-e^{-x}]_0^\infty = 1$. **Converges; value = 1**.
- (f) $\int_2^\infty \frac{dx}{x(\ln x)^2}$. Substitution $t = \ln x$, $dt = dx/x$: $\int_{\ln 2}^\infty t^{-2} dt = [-t^{-1}]_{\ln 2}^\infty = 1/\ln 2$. **Converges; value = 1/\ln 2**.
- (g) $\int_1^\infty (\sqrt{x+1} - \sqrt{x-1}) dx$. Rationalize: $\sqrt{x+1} - \sqrt{x-1} = \frac{2}{\sqrt{x+1} + \sqrt{x-1}} \sim \frac{2}{2\sqrt{x}} = \frac{1}{\sqrt{x}}$. Since $p = 1/2 < 1$, $\int_1^\infty x^{-1/2} dx$ diverges. By limit comparison, **diverges**.
- (h) $\int_0^1 \frac{\ln(1+\sqrt[3]{x})}{e^{\sin x}-1} dx$. Near 0: $\ln(1+\sqrt[3]{x}) \sim x^{1/3}$ and $e^{\sin x} - 1 \sim \sin x \sim x$. So integrand $\sim x^{1/3}/x = x^{-2/3}$. Since $p = 2/3 < 1$, **converges**.
- (i) $\int_0^\infty \frac{\sin x}{x^{3/2}} dx$. Split at $x = 1$:
- Near 0: $\sin x/x^{3/2} \sim x/x^{3/2} = x^{-1/2}$, $p = 1/2 < 1$, convergent.
 - At ∞ : $|\sin x/x^{3/2}| \leq x^{-3/2}$, $p = 3/2 > 1$, convergent absolutely.

Converges.

- (j) $\int_0^1 \frac{e^x-1}{\sin^2 x} dx$. Near 0: $e^x-1 \sim x$ and $\sin^2 x \sim x^2$, so ratio $\sim x^{-1}$. Since $p = 1$, $\int_0^1 x^{-1} dx$ diverges. By limit comparison, **diverges**.
- (k) $\int_1^\infty \frac{\arctan x}{x^2} dx$. Since $\arctan x \leq \pi/2$, we have $\left|\frac{\arctan x}{x^2}\right| \leq \frac{\pi/2}{x^2}$. $\int_1^\infty x^{-2} dx$ converges, so **converges** by comparison.

- (l) $\int_1^\infty (\sqrt{x^2+1} - \sqrt{x^2-1}) dx$. Rationalize: $\sqrt{x^2+1} - \sqrt{x^2-1} = \frac{2}{\sqrt{x^2+1} + \sqrt{x^2-1}} \sim \frac{2}{2x} = \frac{1}{x}$. Since $\int_1^\infty x^{-1} dx$ diverges, **diverges**.
- (m) $\int_0^1 \frac{x^{1/3} \ln x}{1-x^2} dx$. Near 0: $1-x^2 \rightarrow 1$, so integrand $\sim x^{1/3} \ln x \rightarrow 0$ (bounded); no issue at 0. Near 1: let $x = 1-t$, $t \rightarrow 0^+$. $1-x^2 = (1-x)(1+x) \sim 2t$, $x^{1/3} \rightarrow 1$, $\ln x \sim -(1-x) = -t$... wait $\ln(1-t) \sim -t$. So integrand $\sim \frac{1 \cdot (-t)}{2t} = -1/2$. Bounded near 1. **Converges**.

Exercise 2: Topology

- (a) $E = \{x^2 + y^2 < 1\}$ (open disk). **Open, not closed**. Bounded. **Not compact** (not closed). Boundary: the circle $x^2 + y^2 = 1$.
- (b) $E = \{x^2 + y^2 \leq 1\}$ (closed disk). **Closed, not open**. Bounded. **Compact**.
- (c) $E = \{x^2 + y^2 = 1\}$ (unit circle). **Closed, not open**. Bounded. **Compact**. Every point is a boundary point.
- (d) $E = \{x^2 + y^2 + z^2 \leq 4\}$ (closed ball in $\mathbb{R}^3$). **Closed**. Bounded. **Compact**.
- (e) $E = \{1 < x^2 + y^2 \leq 4\}$ (annulus, inner boundary excluded). **Neither open nor closed**. Bounded. **Not compact** (inner circle $x^2 + y^2 = 1$ is limit of points in E but not in E).
- (f) $E = \{y > x^2\}$ (above parabola). **Open** (preimage of open set under continuous map). **Not closed** (e.g., $(0, 0)$ is a limit point not in E). **Unbounded**. **Not compact**.
- (g) $E = \{x > 0, y > 0, xy \leq 1\}$. Contains its boundary (hyperbola $xy = 1$ and parts of axes... actually $x > 0, y > 0$ excludes axes). **Not open** (boundary $xy = 1$ is included). **Not closed** (sequences going to $x \rightarrow 0^+$). **Unbounded**. **Not compact**.
- (h) $E = \{|x| + |y| \leq 1\}$ (diamond/square rotated 45). **Closed**. Bounded. **Compact**.
- (i) $E = \mathbb{Q}^2$. Every neighborhood of every rational point contains irrational points (exterior), so $\mathbb{Q}^2$ has no interior points. Also every real point is a limit of rationals, so the boundary is all of $\mathbb{R}^2$. E is **neither open nor closed**. Unbounded. **Not compact**.
- (j) $E = \{(x, y) : y = \sin(1/x), x > 0\}$ (graph of $\sin(1/x)$). Not open (1-dimensional curve). Is it closed? The closure includes the segment $\{x = 0, -1 \leq y \leq 1\}$ (as $x \rightarrow 0^+$, $\sin(1/x)$ oscillates). So E is **not closed**. Unbounded. **Not compact**.

Exercise 3: Norms and inequalities

- (a) $\bar{x} = (3, -4, 0)$: $\|\bar{x}\|_1 = 3+4+0 = 7$, $\|\bar{x}\|_2 = \sqrt{9+16+0} = 5$, $\|\bar{x}\|_\infty = \max(3, 4, 0) = 4$.
- (b) $\bar{x} = (1, 2, -2, 0)$: $\|\bar{x}\|_1 = 1+2+2+0 = 5$, $\|\bar{x}\|_2 = \sqrt{1+4+4+0} = 3$, $\|\bar{x}\|_\infty = 2$.
- (c) For $\bar{x} = (x_1, x_2) \in \mathbb{R}^2$:

- $\|\bar{x}\|_\infty = \max(|x_1|, |x_2|) \leq \sqrt{x_1^2 + x_2^2} = \|\bar{x}\|_2$: since $\max(|x_1|, |x_2|)^2 \leq x_1^2 + x_2^2$.
 - $\|\bar{x}\|_2 = \sqrt{x_1^2 + x_2^2} \leq |x_1| + |x_2| = \|\bar{x}\|_1$: by $(|x_1| + |x_2|)^2 = x_1^2 + 2|x_1||x_2| + x_2^2 \geq x_1^2 + x_2^2$.
- (d) $\bar{x} = (1, 2, 3)$, $\bar{y} = (4, -1, 2)$. $\bar{x} \cdot \bar{y} = 4 - 2 + 6 = 8$. $|\bar{x}| = \sqrt{14}$, $|\bar{y}| = \sqrt{21}$. C-S says $|\bar{x} \cdot \bar{y}| \leq |\bar{x}||\bar{y}|$: $8 \leq \sqrt{14 \cdot 21} = \sqrt{294} = 7\sqrt{6} \approx 17.1$. ✓
- (e) The set $\|\bar{x}\|_1 \leq 1$ in $\mathbb{R}^2$ is $\{|x_1| + |x_2| \leq 1\}$, which is the square (diamond) with vertices $(\pm 1, 0)$ and $(0, \pm 1)$. The unit disk $\|\bar{x}\|_2 \leq 1$ is inscribed in this diamond (touching at the vertices); the diamond contains the disk.
- (f) **Proof of Cauchy–Schwarz:** For any $\lambda \in \mathbb{R}$,

$$0 \leq \int_0^1 (f(x) - \lambda g(x))^2 dx = \int_0^1 f^2 dx - 2\lambda \int_0^1 fg dx + \lambda^2 \int_0^1 g^2 dx.$$

This is a nonneg quadratic in λ , so its discriminant is ≤ 0 :

$$4 \left(\int_0^1 fg \right)^2 - 4 \left(\int_0^1 f^2 \right) \left(\int_0^1 g^2 \right) \leq 0.$$

Rearranging gives the Cauchy–Schwarz inequality. ✓

Exercise 4: Exact values

- (a) $\int_1^\infty x^{-2} dx = [-x^{-1}]_1^\infty = 1$.
- (b) $\int_0^1 x^{-1/2} dx = [2\sqrt{x}]_0^1 = 2$.
- (c) $\int_1^\infty \frac{\ln x}{x^2} dx$. Integration by parts: $u = \ln x$, $dv = x^{-2} dx$, $du = dx/x$, $v = -x^{-1}$.
 $= [-\frac{\ln x}{x}]_1^\infty + \int_1^\infty \frac{dx}{x^2} = 0 + [-\frac{1}{x}]_1^\infty = 1$.
- (d) $\int_0^\infty xe^{-x} dx$. Integration by parts: $u = x$, $dv = e^{-x} dx$. $= [-xe^{-x}]_0^\infty + \int_0^\infty e^{-x} dx = 0 + 1 = 1$.
- (e) $\int_0^\infty \frac{dx}{(1+x^2)^2}$. Let $x = \tan \theta$, $dx = \sec^2 \theta d\theta$, $1 + x^2 = \sec^2 \theta$. $= \int_0^{\pi/2} \frac{\sec^2 \theta}{\sec^4 \theta} d\theta = \int_0^{\pi/2} \cos^2 \theta d\theta = \int_0^{\pi/2} \frac{1+\cos 2\theta}{2} d\theta = \frac{\pi}{4}$.